

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname _____

Forename(s) _____

Candidate signature _____

I declare this is my own work.

INTERNATIONAL A-LEVEL CHEMISTRY (9620)

Unit 3: Inorganic 2 and Physical 2

Thursday 11 January 2024 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- the Periodic Table/Data Sheet, provided as an insert
- a ruler with millimetre measurements
- a scientific calculator, which you are expected to use where appropriate.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do **not** write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- All working must be shown.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 80.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



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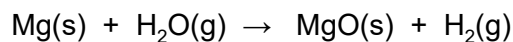
Answer **all** questions in the spaces provided.

0 1

This question is about Period 3 elements and their compounds.

0 1 . 1

Magnesium reacts rapidly with steam in a redox reaction.



Give **two** observations in this reaction.

Explain, by reference to oxidation states, why this is a redox reaction.

[4 marks]

Observation 1 _____

Observation 2 _____

Explanation _____

0 1 . 2

Phosphorus reacts with chlorine to form PCl_5

In the solid state, PCl_5 exists as two ions PCl_4^+ and PCl_6^-

Name the shape of the PCl_4^+ ion.

Give the bond angle in the PCl_4^+ ion.

Draw the structure of PCl_6^- showing clearly the 3D shape of the ion.

[3 marks]

Shape of PCl_4^+ _____

Bond angle in PCl_4^+ _____

Structure of PCl_6^- _____

Turn over ►



0 1 . 3 Write an equation for the reaction of PCl_5 with water.

[1 mark]

0 1 . 4 A Period 3 element reacts with oxygen to form an acidic oxide.
The oxide contains 53% of oxygen by mass.

Identify the Period 3 element.

Tick (✓) **one** box.

[1 mark]

Silicon

☐

Sulfur

☐

Phosphorus

☐

0 1 . 5 Aluminium oxide is amphoteric.
This means that aluminium oxide reacts with acids and with alkalis.

Write **two** ionic equations to show how aluminium oxide is amphoteric.

[2 marks]

Equation 1

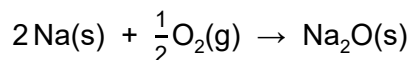
Equation 2



0 2

This question is about thermodynamics.

Sodium reacts with oxygen to form sodium oxide.



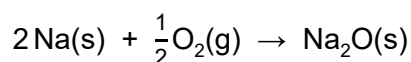
0 2 . 1

Give **two** observations in this reaction.**[2 marks]**

Observation 1 _____

Observation 2 _____

0 2 . 2

The entropy change (ΔS) for this reaction is $-131.7 \text{ J K}^{-1} \text{ mol}^{-1}$ **Table 1** shows some standard entropy values.**Table 1**

	$S^\ominus / \text{J K}^{-1} \text{ mol}^{-1}$
Na(s)	51.0
Na ₂ O(s)	72.8

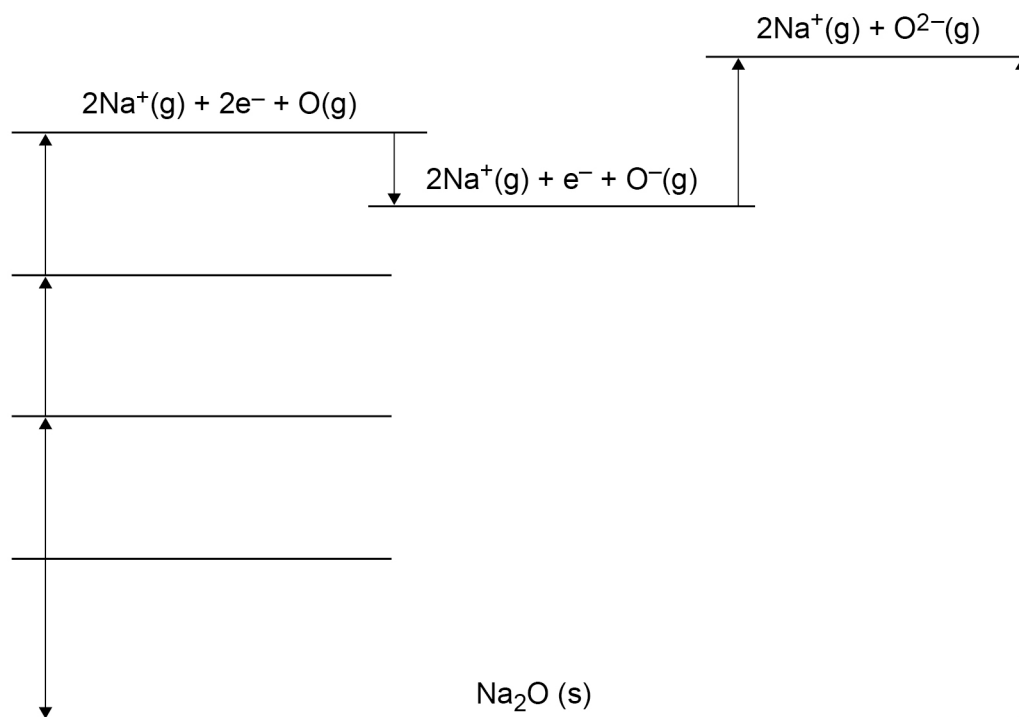
Use the data to calculate a value for the standard entropy for O₂(g)Comment on the difference in the standard entropy values for O₂ and for Na**[4 marks]**Entropy for O₂(g) _____ J K⁻¹ mol⁻¹

Comment _____

Turn over ►

0 2 . 3 **Figure 1** shows an incomplete Born–Haber cycle for the formation of sodium oxide.

Figure 1



Complete **Figure 1** by writing the formulas, including state symbols, of the appropriate species on each of the three horizontal blank lines.

[3 marks]



0 2 . 4 Table 2 shows some enthalpy change data.

Table 2

	Enthalpy change / kJ mol^{-1}
Enthalpy of formation of $\text{Na}_2\text{O}(\text{s})$	−416
Enthalpy of atomisation of sodium	+109
Enthalpy of atomisation of oxygen	+248
First ionisation energy for $\text{Na}(\text{g})$	+494
First electron affinity of oxygen	−142
Second electron affinity of oxygen	+844

Use **Figure 1** and the data in **Table 2** to calculate a value, in kJ mol^{-1} , for the enthalpy of lattice dissociation of $\text{Na}_2\text{O}(\text{s})$

[2 marks]

Enthalpy of lattice dissociation of $\text{Na}_2\text{O}(\text{s})$ _____ kJ mol^{-1}

Question 2 continues on the next page

Turn over ►



0 2 . 5 Table 3 shows some more enthalpy change data.

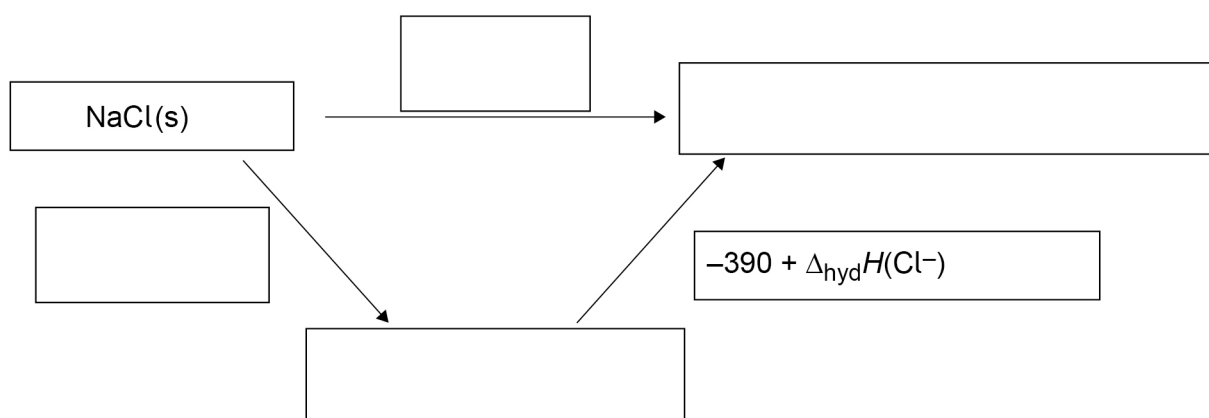
Table 3

	Enthalpy change / kJ mol^{-1}
Enthalpy of lattice dissociation of NaCl(s)	+787
Enthalpy of hydration of $\text{Na}^+(\text{g})$	-390
Enthalpy of hydration of $\text{Cl}^-(\text{g})$	to be calculated
Enthalpy of solution of NaCl(s)	+4

Complete the Hess' Cycle by adding the appropriate species and values.

Use the data in **Table 3** to calculate a value, in kJ mol^{-1} , for the enthalpy of hydration of $\text{Cl}^-(\text{g})$

[2 marks]



Enthalpy of hydration of $\text{Cl}^-(\text{g})$ _____ kJ mol^{-1}

0 2 . 6 Suggest why there is no data book value for the enthalpy of solution of $\text{Na}_2\text{O(s)}$

[1 mark]



Turn over for the next question

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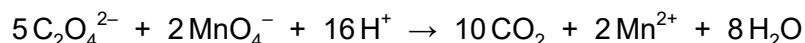
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0 3

This question is about the reaction of ethanedioate ions ($\text{C}_2\text{O}_4^{2-}$) with manganate(VII) ions (MnO_4^-)



The reaction between $\text{C}_2\text{O}_4^{2-}$ and MnO_4^- is slow at the start because the activation energy is high. The reaction then speeds up when some Mn^{2+} ions are formed.

0 3 . 1

Explain why the reaction has a high activation energy.

[1 mark]

0 3 . 2

Write **two** equations to show how Mn^{2+} ions act as a catalyst for this reaction.

[2 marks]

Equation 1

Equation 2

A student completes an experiment to determine whether a solid is lithium ethanedioate, sodium ethanedioate, or potassium ethanedioate.

Method

1. Dissolve 1.775 g of the solid metal ethanedioate in about 100 cm^3 water.
2. Make the solution up to 250 cm^3 with dilute sulfuric acid in a volumetric flask and mix well.
3. Use a pipette to transfer a 25.0 cm^3 portion of the solution into a conical flask.
4. Warm the flask to about 50°C in a water bath.
5. Add $0.0200\text{ mol dm}^{-3}$ potassium manganate(VII) from a burette until there is a permanent colour change.
6. Repeat until two titres are obtained that are within 0.10 cm^3 of each other.

Table 4, on page 11, shows data from the experiment.



Table 4

	Rough	1	2	3
Final burette reading / cm ³	22.40	22.80	44.55	24.65
Initial burette reading / cm ³	0.00	1.50	22.80	3.25
Titre volume / cm ³	22.40	21.30	21.75	21.40

0 3 . 3 Use appropriate results to determine the mean titre for this experiment.

Use the mean titre to calculate the M_r of the solid metal ethanedioate.

Identify the metal ion in the solid.

[6 marks]

Mean titre _____ cm³

M_r of solid metal ethanedioate _____

Metal ion _____

Turn over ►



0 3 . 4 Describe the colour change at the end point of this titration.

Give a reason for this colour change.

[2 marks]

Colour change _____

Reason _____

11

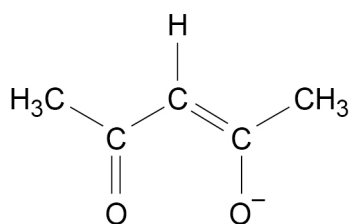
0 4 This question is about bidentate ligands and multidentate ligands.

0 4 . 1 **Figure 2** shows the structure of the **acac** ion ($\text{C}_5\text{H}_7\text{O}_2^-$)
The **acac** ion can act as a bidentate ligand.

Draw **two** lone pairs on the structure to show how the **acac** ion can act as a bidentate ligand.

[1 mark]

Figure 2



acac ion

0 4 . 2 The complex $\text{Co}(\text{C}_5\text{H}_7\text{O}_2)_3$ contains three **acac** ions.

Give the oxidation state of cobalt in this complex.

State the type of isomerism shown by the complex.

[2 marks]

Oxidation state of cobalt _____

Type of isomerism _____



9

0 5

This question is about the reactions of three aqueous solutions (**P**, **Q** and **R**).
The aqueous solutions each contain a different transition metal ion: Fe^{3+} , Fe^{2+} or Cu^{2+}

A student adds three different reagents to the three solutions.

Table 5 shows some observations from the experiments.

Table 5

	Test 1 Add $\text{NH}_3(\text{aq})$ drop by drop until in excess	Test 2 Add $\text{Na}_2\text{CO}_3(\text{aq})$	Test 3 Add concentrated $\text{HCl}(\text{aq})$
P	Blue precipitate formed Blue precipitate reacts to give deep blue solution with excess	Blue–green precipitate formed	Solution turns from blue to yellow–green
Q	Green precipitate formed No further change with excess	Green precipitate formed	
R	Brown precipitate formed No further change with excess	Brown precipitate formed Effervescence	

0 5

1

Write an equation for the reaction in **Test 1** for the formation of the deep blue solution.

[1 mark]



0 5 . 2

The wavelength of the light absorbed by the deep blue solution formed in **Test 1** is 650 nm

Calculate the energy gap (ΔE), in J, between the ground state and the excited state of the d electrons.

You should assume that all the light absorbed is because of the movement of electrons between d orbitals.

The Planck constant, $h = 6.63 \times 10^{-34}$ J s

The speed of light, $c = 3.00 \times 10^8$ m s⁻¹

[2 marks]

ΔE _____ J

0 5 . 3

State the type of reaction that occurs when solution **P** reacts with concentrated HCl

Write an equation for this reaction.

[2 marks]

Type of reaction _____

Equation

0 5 . 4

Identify a reagent, or combination of reagents, used in a test-tube reaction to show that solution **P** also contains sulfate(VI) ions.

Describe the expected observation(s).

[2 marks]

Reagent(s) _____

Observation(s) _____

Turn over ►



Table 5 is repeated here.

Table 5

	Test 1 Add $\text{NH}_3(\text{aq})$ drop by drop until in excess	Test 2 Add $\text{Na}_2\text{CO}_3(\text{aq})$	Test 3 Add concentrated $\text{HCl}(\text{aq})$
P	Blue precipitate formed Blue precipitate reacts to give deep blue solution with excess	Blue–green precipitate formed	Solution turns from blue to yellow–green
Q	Green precipitate formed No further change with excess	Green precipitate formed	
R	Brown precipitate formed No further change with excess	Brown precipitate formed Effervescence	

0 5 . 5 Solution **Q** forms a green precipitate when it reacts in **Test 1**, and a green precipitate when it reacts in **Test 2**.

Give the formula for each green precipitate.

[2 marks]

Formula of green precipitate formed in **Test 1** _____

Formula of green precipitate formed in **Test 2** _____

0 5 . 6 Write an equation for the reaction that occurs when solution **R** reacts in **Test 2**.

[1 mark]



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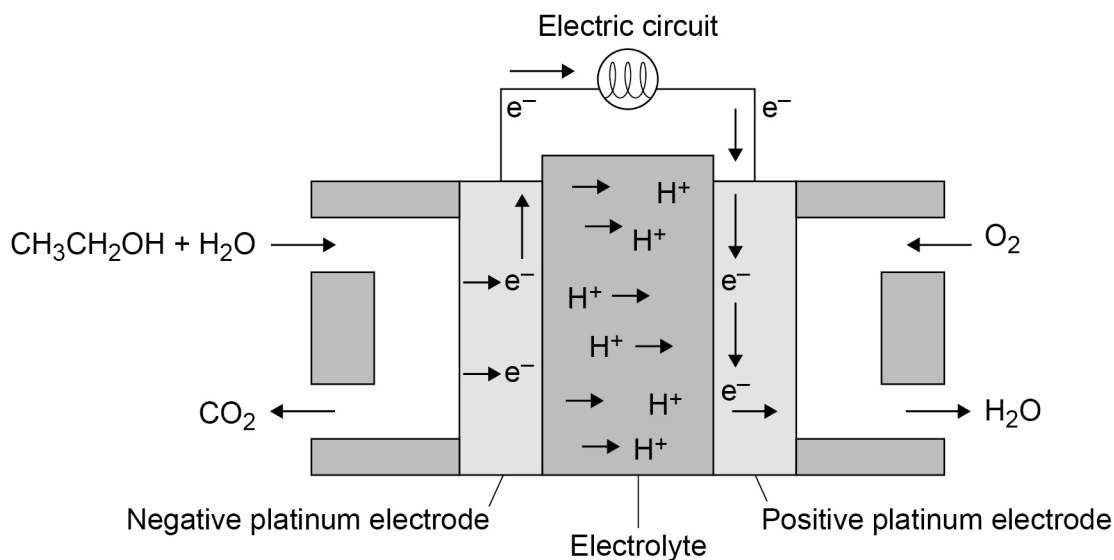


0 6

This question is about fuel cells.

Figure 3 shows an ethanol fuel cell.

Figure 3



When the cell is in use:

- At the positive electrode, oxygen reacts with hydrogen ions to form water.
- At the negative electrode, ethanol reacts with water to form carbon dioxide and hydrogen ions.

0 6 . 1

Deduce the half-equation for the reaction at the positive electrode.

Deduce the half-equation for the reaction at the negative electrode.

Write the equation for the overall reaction that occurs in the ethanol fuel cell.

[3 marks]

Positive electrode

Negative electrode

Overall cell reaction



0 6 . 2 The EMF of the ethanol fuel cell is 1.14 V

The electrode potential for the positive electrode is 1.23 V

Calculate the electrode potential for the negative electrode.

[1 mark]

Electrode potential _____ V

0 6 . 3 State what must be done to maintain a constant EMF during the use of this fuel cell.

[1 mark]

Ethanol fuel cells are being developed as a possible alternative to the alkaline hydrogen–oxygen fuel cell to power vehicles.

In the alkaline hydrogen–oxygen fuel cell, the reactions that occur at each platinum electrode are

Positive platinum electrode $\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightarrow 4\text{OH}^-$

Negative platinum electrode $2\text{H}_2 + 4\text{OH}^- \rightarrow 4\text{H}_2\text{O} + 4\text{e}^-$

0 6 . 4 Give the conventional cell representation for the alkaline hydrogen–oxygen fuel cell.

[2 marks]

0 6 . 5 State **one** disadvantage of the alkaline hydrogen–oxygen fuel cell compared with the ethanol fuel cell.

[1 mark]



07

This question is about acids and bases.

07.1

Define pH

[1 mark]**07.2**

Some concentrated hydrochloric acid contains 38.0 g of HCl in 100 g of acid.
The concentrated acid solution has a density of 1.19 g cm^{-3}

Calculate the concentration, in mol dm^{-3} , of the solution.
Give your answer to three significant figures.

[2 marks]Concentration _____ mol dm^{-3} **07.3**

A teacher needs to make 4.00 dm^3 of a dilute solution of hydrochloric acid with a pH of 0.80 using the concentrated solution.

Use your answer to Question **07.2** to calculate the volume, in cm^3 , of concentrated HCl that is needed to prepare this dilute aqueous solution.

(If you were unable to answer Question **07.2**, you should assume that the concentration of the concentrated HCl is 7.25 mol dm^{-3}
This is **not** the correct answer.)

[3 marks]Volume _____ cm^3 

07.4

1.50 g of barium hydroxide ($M_r = 171.3$) is dissolved in water to give 100 cm³ of aqueous barium hydroxide.

Calculate the pH of this solution at 298 K
Give your answer to two decimal places.

At 298 K the ionic product of water, $K_w = 1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$

[4 marks]

pH _____

07.5

1.50 g of calcium hydroxide is added to water to give 100 cm³ of aqueous calcium hydroxide.

Explain why the pH of this solution is lower than the pH of the barium hydroxide solution prepared in Question **07.4**.

[1 mark]

Question 7 continues on the next page

Turn over ►

A weak acid, HA, has an acid dissociation constant (K_a) of $1.86 \times 10^{-5} \text{ mol dm}^{-3}$ at 298 K

0 7 . 6

Calculate the pK_a for this weak acid at 298 K

Give your answer to two decimal places.

[1 mark]

pK_a _____

0 7 . 7

10.0 cm³ of 0.350 mol dm⁻³ aqueous sodium hydroxide are added to 25.0 cm³ of a 0.250 mol dm⁻³ aqueous solution of HA

Calculate the pH of the solution formed.
Give your answer to two decimal places.

[5 marks]

pH _____

17

END OF QUESTIONS



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